Project: Applying Optical Flow to Detect Moving Objects

In the "Applying Optical Flow" live script reading, `applyingOpticalFlow.mlx`, you used optical flow to identify moving objects with the highest flow vector magnitude.

Here, you will apply optical flow to a few frames taken from a video of a highway.

You will use the flow vectors to create a mask of moving objects. Later, you will use this mask to determine how many cars are traveling in each direction and their velocities.

In this problem, you will:

- Use the Farneback method to calculate the optical flow vectors between the frames and save them as a variable named `flow`.
- Create a mask including only pixels with an optical flow vector magnitude above 1. Save the result as a variable named `mask`.
- Use image processing to update `mask` to remove regions with an area below 500 pixels.
- Use image processing to morphologically close `mask` with a structuring element of type "disk" and size 20.

You will be graded on the following:

- Correct values for each flow vector, `flow`
- A correct mask of the moving objects, `mask`

Script

Save Reset MATLAB Documentation (https://www.mathworks.com/help/)

```
frame1 = imread("Rt9Frame1.png");
frame2 = imread("Rt9Frame2.png");

of = opticalFlowFarneback;
estimateFlow(of,im2gray(frame1));
flow = estimateFlow(of,im2gray(frame2));
vm = flow.Magnitude;
mask = vm(:,:) > 1;
mask = bwareafilt(mask, [500 inf]);
mask = imclose(mask, strel('disk', 20, 0));

% Uncomment below to view your optical flow vectors on your masked image
maskedFrame = frame2; maskedFrame(repmat(~mask, [1 1 3])) = 0;
imshow(frame2)
hold on
plot(flow,"DecimationFactor",[15 15],"ScaleFactor",7)
hold off
figure
imshow(maskedFrame)
```

Run Script

Assessment: All Tests Passed

Submit

Optical Flow Vectors are Correct

The Mask is Correct

Output